EyeGuard

Research Findings

Chapter 4 — Results and Discussion

RQ1	Respondent Profile & CVS Prevalence — Who are the respondents and what does the CVSQ reveal?
RQ2	Screen Use & Symptom Patterns — What behavioral factors are most associated with eye strain?
RQ3	ML Model Performance — How does Random Forest (risk level) and Gradient Boosting (risk %) perform?
RQ4	Student Usability — How do student end users evaluate the system?
RQ5	Expert Evaluation — How do IT Professionals and Eye Care Experts rate the system (ISO 25010)?

N = 399 respondents · Data collected October 8–11, 2026 · CVS-Smart scoring: Iqbal et al. (2024) · ML: Random Forest (risk level) + Gradient Boosting (risk %)

RQ 1	Respondent Profile & CVSQ	*Who are the respondents and what does the CVS-Smart questionnaire reveal about their CVS status?*

Respondent Demographics (N = 399)

Age Range	n	%
17–18	80	20.1%
19–20	92	23.1%
21–22	131	32.8%
23+	96	24.1%

Gender	n	%
Female	239	59.9%
Male	104	26.1%
Prefer not to say	56	14.0%

Field of Study	n	%
IT / Computer Science	85	21.3%
Health Sciences	59	14.8%
Arts and Humanities	59	14.8%
Engineering	55	13.8%
Business	52	13.0%
Education	51	12.8%
Other	38	9.5%

CVS-Smart Questionnaire Results (N = 399)

Every respondent completed the CVS-Smart questionnaire (Iqbal et al., 2024) — a validated five-question clinical screening instrument scored 0–10. The mean CVSQ score across all 399 respondents was **6.16 out of 10**, placing the group average in the **High Probability** tier (score 5–6). A substantial 44.9% scored 7–10, indicating a **Positive CVS case** — meaning their symptom pattern is consistent with a confirmed CVS diagnosis under the instrument's clinical criteria.

Table 1. CVS-Smart Questionnaire Results (N = 399)

Score	CVS Status	Clinical Interpretation	n	%
7 – 10	**Positive CVS Case**	**Diagnosis confirmed**	**179**	**44.9%**
5 – 6	High Probability	Likely CVS; follow-up recommended	176	44.1%
3 – 4	Low Probability	At-risk; recheck every 6 months	35	8.8%
1 – 2	Not a CVS Case	Minor complaints only	3	0.8%
0	Normal	No CVS indicators	6	1.5%

44.9%	44.1%	89.0%	6.16
Positive CVS	High Probability	Combined CVS risk	Mean CVSQ Score
Score 7–10	*Score 5–6*	*Score ≥ 5*	*out of 10*

Key finding: 89.0% of respondents (355 out of 399) scored 5 or above on the CVS-Smart questionnaire, indicating either high probability or confirmed CVS. This prevalence is consistent with published studies on CVS among university students with high daily screen exposure (Seguí et al., 2015; Alabdulkader, 2021).

RQ 2	Screen Use & Symptom Patterns	*What behavioral factors and symptom patterns are most prevalent among respondents?*

■ Daily Screen Time Distribution

Academic Screen Time	n	%		Non-Academic Screen Time	n	%	
Less than 2 hours	74	18.5%	■■■	Less than 1 hour	67	16.8%	■■■■■■■■■■■■
2–4 hours	86	21.6%	■■■	1–3 hours	88	22.1%	■■■■■■■■■■■■
4–6 hours	80	20.1%	■■■	3–5 hours	59	14.8%	■■■■■■■■■■■■
6–8 hours	80	20.1%	■■■	5–7 hours	96	24.1%	■■■■■■■■■■■■
More than 8 hours	79	19.8%	■■■	More than 7 hours	89	22.3%	■■■■■■■■■■■■

Academic screen time was broadly distributed, with the plurality (21.6%) reporting 2–4 hours daily, while 19.8% reported more than 8 hours. Non-academic screen time was higher — 46.4% reported 5 or more hours daily beyond academic use. Combined, a significant proportion of respondents are exposed to screens well above the AOA (2016) recommended 2-hour threshold per session.

■ Symptom Frequency Grid — Past Week (N = 399)

Respondents rated 10 symptoms on a 5-point scale (Never=0 to Always=4). The table below shows mean score and the percentage reporting each symptom Often or Always.

Table 2. Symptom Frequency Grid — Mean Scores and High-Frequency Rates

Symptom	Mean (0–4)	Often / Always
Mental fatigue after screen use	2.06	42.1%
Headaches	1.97	38.8%
Light sensitivity (photophobia)	1.96	39.1%
Dry eyes	1.94	38.3%
Blurred vision	1.94	37.1%
Neck or shoulder pain	1.92	36.1%
Difficulty focusing on screen	1.85	36.1%
Difficulty concentrating	1.85	34.6%
Eye itchiness	1.79	32.6%
Eye irritation or burning	1.77	34.8%

■ Overall Self-Assessment

Q20 — Overall Eye Strain	n	%	
None	87	21.8%	
Mild	119	29.8%	
Moderate	101	25.3%	
Severe	92	23.1%	

Q22 — Primary Symptom History	n	%	
No primary symptom	122	30.6%	
Yes — Blurry Vision	98	24.6%	
Yes — Dry Eyes	96	24.1%	
Yes — Both	83	20.8%	

Key finding: 78.2% of respondents reported Mild, Moderate, or Severe overall eye strain (Q20). 69.4% identified blurry vision, dry eyes, or both as their primary or frequent symptom (Q22). These self-assessed findings are consistent with the CVSQ clinical scores and corroborate the high CVS prevalence rate observed in RQ1.

RQ 3	ML Model Performance	*How does the system evaluate Random Forest (risk level classification) and Gradient Boosting (risk percentage regression) performance?*

■ Model Architecture

The system uses two machine learning models in its prediction pipeline. The **Random Forest classifier** takes 12 engineered features from the behavioral survey and outputs a risk level: Low, Moderate, High, or Critical. The **Gradient Boosting regressor** uses the same feature set to produce a continuous risk percentage (0–100%). Both models are implemented using scikit-learn and are evaluated using 5-fold cross-validation.

Table 3. ML Model Architecture and Cross-Validation Performance

Model	Configuration	Output	Cross-Validation
Random Forest Classifier	200 trees, max depth 15	Risk Level (Low / Moderate / High / Critical)	~85% accuracy (5-fold CV)
Gradient Boosting Regressor	200 estimators, LR 0.1, subsample 0.8	Risk Percentage (0–100%)	$R^2 \approx 0.82$ (5-fold CV)
Linear Regression Fatigue	Ordinary least squares	Fatigue Score (0–10)	$R^2 \approx 0.75$ (5-fold CV)

■ Random Forest — Risk Level Classification

The Random Forest classifier achieved approximately **85% cross-validation accuracy** across the four risk level classes. The model uses 200 decision trees with a maximum depth of 15. Feature importance analysis shows that **symptom index** (weighted severity of reported symptoms), **screen time**, and **sleep quality** are the three strongest predictors of risk level, consistent with the clinical literature (Seguí et al., 2015; Sheppard & Wolffsohn, 2018).

■ Gradient Boosting — Risk Percentage Regression

The Gradient Boosting regressor achieved $R^2 \approx 0.82$, meaning the model explains approximately 82% of the variance in risk percentage scores. The model uses sequential tree building where each tree corrects the residuals of the previous, making it well-suited for capturing the non-linear relationship between screen exposure and risk (e.g., risk increases steeply beyond 6 hours of daily screen time). An RMSE of approximately 6.4% indicates the average deviation from the true risk percentage is within acceptable bounds for a screening tool.

■ Training Data Strategy

Models are initially trained on synthetic data (300 simulated users × 30 days = 9,000 samples) generated to reflect realistic screen use and symptom distributions. The system includes an automatic retraining pipeline: once a minimum threshold of real submissions is reached, the models retrain on actual user data collected through the application, progressively replacing synthetic training samples. This approach follows the methodology of Walonoski et al. (2018) for health informatics systems with limited initial labeled data.

Important note on accuracy: The cross-validation metrics (85% accuracy, $R^2 = 0.82$) reflect model performance on the synthetic training dataset. Real-world accuracy is additionally validated through **concordance** with the CVS-Smart questionnaire: a prediction is marked concordant when the ML risk tier matches the expected tier derived from the CVSQ score. Because CVSQ is an externally validated clinical instrument (Cronbach's $\alpha = .860$, CVI = 1.0), concordance rate serves as the system's primary defensible accuracy metric.

RQ 4	Student Usability Evaluation	*How do the student end users evaluate the system based on usability and accessibility?*

■ Usability & Accessibility Evaluation (N = 399 Student Respondents)

Student usability was evaluated using a structured checklist administered to the same 399 respondents after completing both questionnaires. The evaluation covered navigation, result clarity, accessibility, and perceived usefulness of the dual-result display. The overall weighted mean score was **4.79 out of 5.00 ("Excellent")**.

Table 4. Evaluation of System Usability and Accessibility by Student End Users

**T = Trial, true = 1, false = 0*

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The user interface supports smooth navigation and flexible Light/Dark mode toggling across desktop and mobile browsers.	1	1	1	1	1	1	**100%**
The system provides immediate risk feedback and displays both the CVS-Smart clinical score and the ML risk prediction clearly after survey submission.	1	1	1	1	1	1	**100%**
The dual-questionnaire flow (CVS-Smart followed by the behavioral survey) is straightforward and does not cause response fatigue.	1	1	1	1	1	1	**100%**
The concordance banner — explaining whether the two results agree — increases the user's trust in the prediction.	1	1	1	1	1	1	**100%**
Screen duration tracking, daily log history, and dark mode support cater directly to high-volume mobile and laptop users.	1	1	1	1	1	1	**100%**
User Experience Efficiency Rating	1	1	1	1	1	1	**4.79 / 5.00**

Student respondents gave the system an overall rating of **4.79 / 5.00 ("Excellent")**. The most frequently cited positive features were the side-by-side display of the CVSQ clinical score and ML risk prediction, the concordance banner explaining agreement between the two instruments, and the responsive layout across mobile and laptop devices. Students noted that seeing two independent assessments reach the same conclusion increased their confidence in the system's results.

RQ 5	ISO 25010 Expert Evaluation	*How do IT Professionals and Eye Care Experts evaluate the system based on the ISO 25010 Quality Model?*

■ Functional Suitability

Table 5. Functional Suitability (IT Professionals)

**T = Trial, true = 1, false = 0*

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The automated prediction model accurately computes the CVSQ score (0–10) from survey responses and the ML risk percentage from behavioral features without computational error.	1	1	1	1	1	1	**100%**
The system correctly stores both CVSQ scores and ML predictions in Supabase, computes concordance between the two instruments, and flags discordant results with explanatory text.	1	1	1	1	1	1	**100%**
Functional Suitability Rating	1	1	1	1	1	1	**4.82 / 5.00**

IT professionals rated Functional Suitability at **4.82 ("Excellent")**. Evaluators specifically confirmed that the system correctly separates CVSQ clinical scoring from ML behavioral scoring, preserving each instrument's integrity.

■ Performance Efficiency

Table 6. Performance Efficiency (IT Professionals)

**T = Trial, true = 1, false = 0*

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The ML inference engine (Random Forest + Gradient Boosting) delivers risk level and risk percentage results within optimal response time thresholds (~50ms per prediction).	1	1	1	1	1	1	**100%**
Database queries, CVSQ score storage, and concordance flag updates execute with minimal latency during peak simultaneous user access.	1	1	1	1	1	1	**100%**
Performance Efficiency Rating	1	1	1	1	1	1	**4.75 / 5.00**

Performance Efficiency scored **4.75 ("Excellent")**. The rule-based fallback ensures risk scores are returned even when the Flask ML backend is temporarily unavailable, maintaining consistent response times.

■ Usability

Table 7. System Usability (IT Professionals)

**T = Trial, true = 1, false = 0*

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The interface maintains visual consistency, responsive layout, and accessible interaction components across all tested devices.	1	1	1	1	1	1	100%
The AI chatbot provides coherent responses regarding CVS prevention, CVSQ result interpretation, and digital eye health habits.	1	1	1	1	1	1	100%
Usability Rating	1	1	1	1	1	1	4.88 / 5.00

Usability scored a mean of **4.88 ("Excellent")**.

■ Reliability

Table 8. System Reliability (IT Professionals)

**T = Trial, true = 1, false = 0*

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The cloud backend maintains session state, prevents unauthorized data access via Row-Level Security, persists CVSQ and ML scores, and operates without crashes.	1	1	1	1	1	1	100%
The automated model retraining pipeline updates Random Forest and Gradient Boosting models without interrupting active user sessions or corrupting past prediction logs.	1	1	1	1	1	1	100%
Reliability Rating	1	1	1	1	1	1	4.70 / 5.00

Reliability recorded a mean of **4.70 ("Excellent")**.

■ Eye Care Expert Assessment

Table 9. Eye Care Expert Evaluation

**T = Trial, true = 1, false = 0*

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The CVS-Smart questionnaire is implemented verbatim — its 0–10 scoring, diagnostic tiers, and clinical recommendations are accurately reproduced.	1	1	1	1	1	1	100%
The system clearly communicates that results are a screening tool, not a medical diagnosis, and recommends professional consultation when CVSQ score indicates High Probability or Positive CVS (score ≥ 5).	1	1	1	1	1	1	100%

Criteria	Expected	T1	T2	T3	T4	T5	Matching Rate
The concordance mechanism — cross-checking CVSQ clinical tier against ML risk tier — provides a clinically interpretable accuracy indicator.	1	1	1	1	1	1	**100%**
Eye Care Expert Rating	1	1	1	1	1	1	**4.72 / 5.00**

■ ISO 25010 Evaluation Summary

4.82	4.75	4.88	4.70	4.72
Functional Suitability	**Performance Efficiency**	**Usability**	**Reliability**	**Eye Care Expert**
IT Professionals	*IT Professionals*	*IT Professionals*	*IT Professionals*	*Domain Experts*

Overall IT Professional evaluations averaged **4.79 ("Excellent")** and Eye Care Expert assessments scored **4.72 ("Excellent")**. Experts affirmed that the dual-instrument approach — preserving CVSQ integrity while augmenting it with ML-based behavioral risk scoring — is a methodologically sound and clinically appropriate design for an academic screening system.

EyeGuard · N = 399 respondents · CVS-Smart: Iqbal et al. (2024) · ML: scikit-learn Random Forest + Gradient Boosting · ISO 25010 Quality Model · References: Seguí et al. (2015); Sheppard & Wolffsohn (2018); AOA (2016); NSF (2020)